

Enterprise AI Platform, Data & Agentic Architecture

One paved road for every AI workload

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1. Architecture Principles

- **Model-agnostic by construction.** All model access flows through a gateway; frontier models are swappable commodities and no use case binds directly to a vendor SDK.
- **Paved road, not police road.** The platform must be the easiest way to ship an AI feature - golden paths, templates, and self-service - or teams will route around it.
- **Permissions travel with data.** Source-system entitlements are enforced at retrieval time; an AI system must never widen a user's effective access.
- **Everything observable, everything evaluated.** No production AI without logging, tracing, evaluation suites, and cost attribution.
- **Autonomy is earned, not granted.** Agents progress through defined autonomy levels based on demonstrated evaluation performance (Section 4).
- **Build the scarce, buy the commodity.** Buy models, vector stores, and orchestration primitives; build the knowledge corpus, evaluation assets, domain agents, and integration fabric - that is where advantage lives.

2. Reference Platform Blueprint

Layer	Capabilities	Approach & key decisions
7. Experience	Enterprise assistant, in-app copilots, API/embedded AI for products	One assistant surface with function-specific skills; copilots embedded where work happens (IDE, CRM, ITSM)
6. Agent orchestration	Agent runtime, multi-agent collaboration, planner/executor patterns, workflow engine, human-in-the-loop gates, agent registry	Buy orchestration framework; build domain agents. Every agent registered with owner, scope, autonomy level, and tool entitlements
5. Tool & integration fabric	Model Context Protocol (MCP) servers over enterprise systems; A2A patterns for cross-domain agent cooperation; API management	Standardize on MCP as the enterprise tool-integration contract: one governed MCP server per system-of-record (CRM, ERP, ITSM, HRIS, KB), owned by that system's team, with scoped authorization
4. Knowledge & memory	RAG pipelines, vector databases, hybrid search (semantic + keyword), knowledge graph for entities, conversation and task memory with retention policy	Central retrieval service consumed by all use cases; document-level ACLs enforced at query time; memory tiered (session, task, long-term) with privacy-driven TTLs
3. Model layer	Frontier LLMs (multi-vendor), specialized/small models for cost-sensitive tasks, embeddings, fine-tuned adapters where justified, classical ML	AI gateway provides routing, failover, caching, rate limits, and per-team cost metering; model choice per task class is a routing policy, not a code change
2. Data platform	Lakehouse, streaming, data products, semantic layer, feature store, data quality & lineage	Detailed in Section 3
1. Infrastructure	Primary hyperscaler for the platform core; secondary cloud for resilience and model diversity; GPU capacity strategy; on-prem integration	Multi-cloud pragmatism: portable at the gateway and data layers, not force-fitted everywhere; capacity reserved for predictable inference, burst for training/experimentation

Layer	Capabilities	Approach & key decisions
0. Cross-cutting	Identity (human + non-human), secrets, network segmentation, observability, evaluation service, FinOps, CI/CD for agents and prompts	Detailed in Section 5 and Deliverable 04

3. Data & Knowledge Architecture

From data estate to decision fuel

Component	Purpose	Design notes
Data products	Governed, contract-backed datasets owned by domains (customer, product, finance, supply)	Each has an owner, SLA, quality metrics, and consumers; funded as products, not projects
Semantic layer	One set of definitions for core metrics and entities	Prerequisite for trustworthy decision support; Horizon 1 deliverable scoped to ~50 core metrics
Unstructured knowledge pipeline	Ingest, chunk, enrich, and index documents, wikis, tickets, emails (policy-permitting), and call transcripts	Metadata and ACL capture at ingestion; freshness SLAs; deletion propagation to honor privacy requests end-to-end
Knowledge graph	Entities and relationships (customers, products, contracts, assets) that ground retrieval and agent reasoning	Start narrow (customer 360 + product) and grow by use-case pull, not big-bang modeling
Feature store & ML data	Reusable features for classical ML (forecasting, risk scoring)	Shared with LLM workloads where hybrid patterns apply
Knowledge engineering program	Systematic capture of expert know-how (interviews, process mining, annotation) into retrievable and evaluable form	Treat expert time as the scarcest input; prioritize domains with retirement-driven knowledge risk

4. Agentic AI Strategy

Graduated autonomy model

Level	Description	Human role	Example gates to advance
A0 - Assist	Drafts, summarizes, suggests; no system writes	Human performs all actions	Baseline evaluation suite passed
A1 - Act with approval	Agent prepares transactions; human approves each	Human-in-the-loop per action	≥95% suggestion acceptance in pilot; zero critical errors over defined volume
A2 - Act with oversight	Agent executes within hard limits; humans review samples and exceptions	Human-on-the-loop	Sustained A1 performance; rollback tested; limits (value, scope, rate) encoded
A3 - Autonomous in bounds	End-to-end execution in a narrow, reversible domain	Human sets policy, audits outcomes	Independent risk sign-off; kill switch and circuit breakers proven in game-day exercise

Multi-agent and interoperability standards

- **MCP as the tool contract.** Agents never integrate point-to-point with systems; they call governed MCP servers whose scopes are permissioned per agent identity. This turns integration sprawl into a managed catalog.
- **Agent-to-agent (A2A) collaboration** is permitted only between registered agents, with typed task contracts, budget limits (tokens, time, spend), and full trace propagation so a cross-agent workflow is auditable end to end.
- **Orchestration patterns.** Prefer supervisor/worker and pipeline patterns with explicit checkpoints over free-form agent swarms; determinism at the workflow layer, flexibility inside each step.
- **Every agent has an owner.** A named human accountable for its scope, entitlements, evaluation results, and retirement - no orphan agents.

5. Platform Engineering, Observability, Evaluation, and FinOps

Discipline	Minimum standard at scale
Golden paths	Templates for the top patterns (RAG app, copilot, A1 agent) that ship with logging, evals, security, and cost metering pre-wired; team-to-production in days
Observability	Full tracing of prompts, retrievals, tool calls, and outputs; PII-aware log redaction; drift and quality dashboards per use case
Evaluation service	Central eval harness: golden datasets per use case, automated regression on every model/prompt change, red-team suites for safety, human review queues for calibration
LLMOps / CI-CD	Prompts, agent configs, and eval suites are versioned artifacts; promotion between environments gated on eval scores, not vibes
FinOps for AI	Per-use-case token and GPU cost attribution; routing policies that downshift to smaller models where quality permits; caching; monthly unit-economics review (cost per resolved ticket, per PR, per document)
Resilience	Multi-model failover at the gateway; degraded-mode design (fall back to search/human) so AI outage never equals business outage; capacity and quota management

The platform's success metric is not its feature list. It is the median time for a business team to go from idea to governed production deployment - target under 30 days by month 18.